

DRUMHELLER EAST, AB, Canada

WMO: 712370

Lat: 51.450N

Lon: 112.680W

Elev: 2225

StdP: 13.55

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.6	(c) -18.1	(d) -32.6	(e) 0.9	(f) -25.5	(g) -24.7	(h) 1.5	(i) -17.9	(j) 21.0	(k) 26.2	(l) 18.8	(m) 24.8	(n) 2.5	(o) 160	(p) 0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 25.2	(c) 88.8	(d) 62.8	(e) 85.1	(f) 61.6	(g) 81.6	(h) 60.6	(i) 66.2	(j) 81.2	(k) 64.3	(l) 79.8	(m) 62.5	(n) 76.8	(o) 7.7	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.1	87.3	71.7	58.8	80.2	69.4	56.8	74.7	67.6	32.1	81.0	30.6	80.0	29.2	76.8	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years					
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	16.7	14.4	DB	-32.6	94.6	7.0	3.4	-37.6	97.0	-41.7	99.0	-45.6	100.8	-50.6	103.3
			WB	-32.8	69.4	6.9	2.4	-37.8	71.2	-41.8	72.6	-45.7	74.0	-50.7	75.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	11.3	15.9	27.6	41.8	52.9	60.3	66.0	64.2	55.0	41.8	25.8	13.5
	DBStd	22.18	15.40	13.92	14.31	9.04	7.81	5.77	5.46	5.99	7.81	8.52	13.02	14.10
	HDD50	5349	1201	955	694	267	59	2	0	1	38	276	726	1130
	HDD65	9381	1666	1375	1158	696	380	163	54	88	311	720	1175	1595
	CDD50	1626	0	0	1	21	149	310	496	440	187	21	1	0
	CDD65	184	0	0	0	0	5	22	84	63	10	0	0	0
	CDH74	3293	0	0	2	15	206	397	1222	1076	359	16	0	0
	CDH80	1074	0	0	0	2	44	89	426	398	114	1	0	0
	WSAvg	5.8	4.6	4.8	6.0	7.2	7.3	6.8	6.2	5.7	5.7	5.8	5.2	4.6
	PrecAvg	15.1	0.6	0.4	0.6	1.0	1.9	3.0	2.6	2.1	1.3	0.6	0.6	0.5
Precipitation	PrecMax	19.3	1.3	0.9	2.1	2.4	3.7	6.0	5.9	4.3	3.2	1.4	1.9	1.1
	PrecMin	9.0	0.0	0.0	0.1	0.1	0.2	1.3	0.9	0.1	0.2	0.2	0.1	0.0
	PrecStd	2.6	0.3	0.2	0.4	0.6	0.9	1.2	1.2	1.1	0.7	0.4	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.7	49.4	63.5	75.6	85.3	87.3	92.7	93.5	89.4	75.9	60.7	50.8
		MCWB	39.0	39.4	46.0	52.4	57.4	62.7	64.8	63.2	60.4	54.1	46.1	39.8
	2%	DB	43.5	43.9	56.5	70.0	79.6	82.3	88.9	88.7	83.2	67.6	53.5	44.1
		MCWB	36.9	36.3	42.9	49.5	55.7	60.5	64.2	62.3	58.3	50.4	42.0	36.7
	5%	DB	38.7	39.8	51.5	64.3	74.9	78.5	84.8	84.5	77.6	62.3	47.8	38.8
		MCWB	33.5	33.9	40.2	46.5	53.7	58.4	63.2	61.3	56.6	47.7	38.9	33.2
	10%	DB	32.5	35.2	46.4	58.9	70.3	75.0	81.0	80.3	72.3	57.0	42.5	33.3
		MCWB	29.1	30.8	37.8	43.9	51.5	57.1	62.2	60.4	54.5	45.0	35.9	29.6
	0.4%	WB	39.5	40.3	47.4	53.3	59.6	65.8	70.0	67.7	62.1	55.3	46.7	41.4
		MCDB	46.5	48.4	60.9	73.6	79.1	81.2	84.8	83.3	83.0	73.4	59.6	49.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.0	36.8	43.8	50.2	57.2	62.8	67.2	65.4	59.7	51.3	42.9	36.6
		MCDB	43.2	43.0	55.2	67.6	74.9	76.7	82.2	81.8	78.9	65.2	52.5	43.4
	5%	WB	33.6	33.9	40.8	47.4	55.0	60.6	65.1	63.5	57.6	48.6	39.5	33.2
		MCDB	38.4	39.2	50.7	62.6	71.1	74.0	79.8	79.4	75.0	60.2	46.7	38.5
	10%	WB	29.2	31.0	38.0	44.7	52.9	58.5	63.4	61.8	55.3	46.0	36.2	29.7
		MCDB	32.3	35.2	45.9	57.3	67.9	71.5	77.7	76.4	70.9	55.9	41.9	33.5
	5% DB	MDBR	21.1	23.8	22.8	24.3	24.8	22.9	25.2	26.6	27.2	24.6	20.5	20.5
		MCDBR	26.8	28.5	30.1	35.2	33.6	30.0	32.3	35.5	38.0	34.0	27.3	26.2
		MCWBR	20.7	21.2	17.8	18.1	15.0	12.6	12.6	14.0	16.7	18.3	17.2	19.7
		MCDBR	26.2	27.5	29.1	32.2	29.9	26.0	28.3	30.4	36.0	31.9	25.6	25.5
Clear-Sky Solar Irradiance	taub		0.272	0.278	0.292	0.351	0.377	0.370	0.367	0.386	0.334	0.290	0.260	0.248
		taud	2.330	2.401	2.446	2.293	2.271	2.344	2.378	2.305	2.418	2.520	2.475	2.371
	Ebn at Noon		239	273	291	281	278	280	279	267	271	266	247	232
		Edh at Noon	20	24	28	37	40	38	36	36	29	22	17	16
All-Sky Solar Radiation	RadAvg		350	675	1111	1505	1821	1953	2088	1689	1189	718	384	275
	RadStd		26	56	95	117	121	117	103	119	124	84	31	27

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page